

Discrete Probability

Chapter 7

With Question/Answer Animations

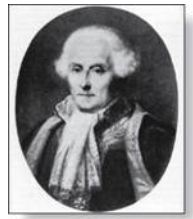
An Introduction to Discrete Probability

Section 7.1



7.1.2 Finite Probability

Probability of an Event₁



Pierre-Simon Laplace
(1749-1827)

We first study Pierre-Simon Laplace's classical theory of probability, which he introduced in the 18th century, when he analyzed games of chance.

We first define these key terms:

- An *experiment* is a procedure that yields one of a given set of possible outcomes.
- The *sample space* of the experiment is the set of possible outcomes.
- An *event* is a subset of the sample space.

Here is how Laplace defined the probability of an event:

Definition: If S is a finite sample space of *equally likely outcomes*, and E is an event, that is, a subset of S , then the *probability of E* is $p(E) = |E|/|S|$.

For every event E , we have $0 \leq p(E) \leq 1$. This follows directly from the definition because $0 \leq p(E) = |E|/|S| \leq |S|/|S| \leq 1$, since $0 \leq |E| \leq |S|$.

Applying Laplace's Definition₁

An experiment: Roll a dice

The sample space: $\{1,2,3,4,5,6\}$

An event E: the result number is even $\{2,4,6\}$

The probability of E : $P(E)=3/6=1/2$

Example: What is the probability that when two dice are rolled, the sum of the numbers on the two dice is 7?

Solution: By the product rule there are $6^2 = 36$ possible outcomes. Six of these sum to 7. Hence, the probability of obtaining a 7 is $6/36 = 1/6$.

Applying Laplace's Definition₃

Example: There are many lotteries that award prizes to people who correctly choose a set of six numbers out of the first n positive integers, where n is usually between 30 and 60. What is the probability that a person picks the correct six numbers out of 40?

Solution: The number of ways to choose six numbers out of 40 is

$$C(40,6) = 40!/(34!6!) = 3,838,380.$$

Hence, the probability of picking a winning combination is $1/3,838,380 \approx 0.00000026$.

Applying Laplace's Definition₄

Example: What is the probability that the numbers 11, 4, 17, 39, and 23 are drawn in that order from a bin with 50 balls labeled with the numbers 1,2, ..., 50 if

- a) The ball selected is not returned to the bin.
- b) The ball selected is returned to the bin before the next ball is selected.

Solution: Use the product rule in each case.

- a) *Sampling without replacement:* The probability is $1/254,251,200$ since there are $50 \cdot 49 \cdot 48 \cdot 47 \cdot 46 = 254,251,200$ ways to choose the five balls.
- b) *Sampling with replacement:* The probability is $1/50^5 = 1/312,500,000$ since $50^5 = 312,500,000$.

7.1.3 The Probability of Complements and Unions of Events₁

Theorem 1: Let E be an event in sample space S . The probability of the event $\overline{E} = S - E$, the complementary event of E , is given by

$$p(\overline{E}) = 1 - p(E).$$

Proof: Using the fact that $|\overline{E}| = |S| - |E|$,

$$p(\overline{E}) = \frac{|S| - |E|}{|S|} = 1 - \frac{|E|}{|S|} = 1 - p(E).$$

The Probability of Complements and Unions of Events₂

Example: A sequence of 10 bits is chosen randomly. What is the probability that at least one of these bits is 0?

Solution: Let E be the event that at least one of the 10 bits is 0. Then $\bar{E}$ is the event that all of the bits are 1s. The size of the sample space S is 2^{10} . Hence,

$$p(E) = 1 - p(\bar{E}) = 1 - \frac{|\bar{E}|}{|S|} = 1 - \frac{1}{2^{10}} = 1 - \frac{1}{1024} = \frac{1023}{1024}.$$

The Probability of Complements and Unions of Events₃

Theorem 2: Let E_1 and E_2 be events in the sample space S . Then

$$p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2)$$

Proof: Given the inclusion-exclusion formula from Section 2.2, $|A \cup B| = |A| + |B| - |A \cap B|$, it follows that

$$\begin{aligned} p(E_1 \cup E_2) &= \frac{|E_1 \cup E_2|}{|S|} = \frac{|E_1| + |E_2| - |E_1 \cap E_2|}{|S|} \\ &= \frac{|E_1|}{|S|} + \frac{|E_2|}{|S|} - \frac{|E_1 \cap E_2|}{|S|} \\ &= p(E_1) + p(E_2) - p(E_1 \cap E_2). \end{aligned}$$

The Probability of Complements and Unions of Events₄

Example: What is the probability that a positive integer selected at random from the set of positive integers not exceeding 100 is divisible by either 2 or 5?

Solution: Let E_1 be the event that the integer is divisible by 2 and E_2 be the event that it is divisible 5. Then the event that the integer is divisible by 2 or 5 is $E_1 \cup E_2$ and $E_1 \cap E_2$ is the event that it is divisible by 2 and 5.

It follows that:

$$\begin{aligned} p(E_1 \cup E_2) &= p(E_1) + P(E_2) - p(E_1 \cap E_2) \\ &= 50/100 + 20/100 - 10/100 = 3/5. \end{aligned}$$

Section Summary₁

Finite Probability: If S is a finite sample space of **equally likely outcomes**, and E is an event, that is, a subset of S , then the **probability of E** is $p(E) = |E|/|S|$.

Probabilities of Complements and Unions of Events

$$p(\overline{E}) = 1 - p(E).$$

$$p(E_1 \cup E_2) = p(E_1) + p(E_2) - p(E_1 \cap E_2)$$

Exercise

20. What is the probability that a five-card poker hand contains a royal flush, that is, the 10, jack, queen, king, and ace of one suit?

扑克牌的四种花色:heart 红桃

spade 黑桃

club 梅花

diamond 方块

扑克牌的名称:

大小王统称joker,

从二到十: two,three,four,five,six,seven,eight,nine,ten.

J是Jack Q是Queen K是King A是Ace

20. There are 4 royal flushes, one in each suit. Therefore the answer is $4/C(52, 5) = 4/2598960 = 1/649740$.

12. What is the probability that a five-card poker hand contains exactly one ace?
13. What is the probability that a five-card poker hand contains at least one ace?
14. What is the probability that a five-card poker hand contains cards of five different kinds?

12, $\frac{4C(48, 4)}{C(52, 5)} \approx 0.30.$

13. $1 - [C(48, 5)/C(52, 5)]$

14. We saw in Example 11 of Section 6.3 that there are $C(52, 5) = 2,598,960$ different hands, and we assume by symmetry that they are all equally likely. We need to count the number of hands that have 5 different kinds (ranks). There are $C(13, 5)$ ways to choose the kinds. For each card, there are then 4 ways to choose the suit. Therefore there are $C(13, 5) \cdot 4^5 = 1,317,888$ ways to choose the hand. Thus the probability is $1317888/2598960 = 2112/4165 \approx 0.51$.

Expected Value and Variance

Section 7.4

7.4.2 Expected Value₁

Definition: The *expected value* (or *expectation*(期望) or *mean*(均值)) of the random variable $X(s)$ on the sample space S is equal to

$$E(X) = \sum_{s \in S} p(s)X(s)$$

Example-Expected Value of a Die: Let X be the number that comes up when a fair die is rolled. What is the expected value of X ?

Solution: The random variable X takes the values 1, 2, 3, 4, 5, or 6. Each has probability $1/6$. It follows that

$$E(X) = \frac{1}{6} \cdot 1 + \frac{1}{6} \cdot 2 + \cdots + \frac{1}{6} \cdot 6 = \frac{21}{6} = \frac{7}{2}.$$

Expected Value₂

Theorem 1: If X is a random variable and $p(X = r)$ is the probability that $X = r$, so that

$$p(X = r) = \sum_{s \in S, X(s)=r} p(s), \text{ then}$$

$$E(X) = \sum_{r \in X(S)} p(X = r)r.$$

Proof: Suppose that X is a random variable with range $X(S)$ and let $p(X = r)$ be the probability that X takes the value r . Consequently, $p(X = r)$ is the sum of the probabilities of the outcomes s such that $X(s) = r$. Hence,

$$E(X) = \sum_{r \in X(S)} p(X = r)r$$

Expected Value₃

Example 3: What is the expected value of the sum of the numbers that appear when a pair of fair dice is rolled?

Solution: Let X be the random variable equal to the sum of the numbers that appear when a pair of dice is rolled. We listed the value of X for the 36 outcomes of this experiment. The range of X is $\{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$.

$$p(X = 2) = p(X = 12) = 1/36,$$

$$p(X = 3) = p(X = 11) = 2/36 = 1/18,$$

$$p(X = 4) = p(X = 10) = 3/36 = 1/12,$$

$$p(X = 5) = p(X = 9) = 4/36 = 1/9,$$

$$p(X = 6) = p(X = 8) = 5/36,$$

$$p(X = 7) = 6/36 = 1/6.$$

Substituting these values in the formula, we have

$$\begin{aligned} E(X) &= 2 \cdot \frac{1}{36} + 3 \cdot \frac{1}{18} + 4 \cdot \frac{1}{12} + 5 \cdot \frac{1}{9} + 6 \cdot \frac{5}{36} + 7 \cdot \frac{1}{6} \\ &\quad + 8 \cdot \frac{5}{36} + 9 \cdot \frac{1}{9} + 10 \cdot \frac{1}{12} + 11 \cdot \frac{1}{18} + 12 \cdot \frac{1}{36} \\ &= 7. \end{aligned}$$

7.2.6 Bernoulli Trials and the Binomial Distribution

Bernoulli Trials₁

James Bernoulli
(1654 – 1705)



Definition: Suppose an experiment can have only two possible outcomes, *e.g.*, the flipping of a coin or the random generation of a bit.

- Each performance of the experiment is called a *Bernoulli trial*.
- One outcome is called a *success* and the other a *failure*.
- If p is the probability of *success* and q the probability of *failure*, then $p + q = 1$.

Many problems can be solved by determining the probability of k successes when an experiment consists of n *mutually independent* Bernoulli trials.

Bernoulli Trials₂

Example: A coin is biased so that the probability of heads is $2/3$. What is the probability that exactly four heads occur when the coin is flipped seven times?

Solution: There are $2^7 = 128$ possible outcomes. The number of ways four of the seven flips can be heads is $C(7,4)$. The probability of each of the outcomes is $(2/3)^4(1/3)^3$ since the seven flips are independent. Hence, the probability that exactly four heads occur is

$$C(7,4) (2/3)^4(1/3)^3 = (35 \cdot 16) / 3^7 = 560 / 2187.$$

Probability of k Successes in n Independent Bernoulli Trials.

Theorem 2: The probability of exactly k successes in n independent Bernoulli trials, with probability of success p and probability of failure $q = 1 - p$, is

$$C(n,k)p^kq^{n-k}.$$

Proof: The outcome of n Bernoulli trials is an n -tuple $(t_1, t_2, \dots, t_n)$, where each is t_i either S (success) or F (failure). The probability of each outcome of n trials consisting of k successes and $n-k$ failures (in any order) is p^kq^{n-k} . Because there are $C(n,k)$ n -tuples of S s and F s that contain exactly k S s, the probability of k successes is $C(n,k)p^kq^{n-k}$.

Expected Value₃

Theorem 2: The expected number of successes when n mutually independent Bernoulli trials are performed, where, p is the probability of success on each trial, *is np .*

7.4.3 Linearity of Expectations₁

The following theorem tells us that **expected values are linear**. For example, the expected value of the sum of random variables is the sum of their expected values.

Theorem 3: If X_i , $i = 1, 2, \dots, n$ with n a positive integer, are random variables on S , and if a and b are real numbers, then

$$(i) \quad E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n)$$

$$(ii) \quad E(aX + b) = aE(X) + b.$$

Linearity of Expectations_{2c}

Example 4: Use Theorem 3 to find the expected value of the sum of the numbers that appear when a pair of fair dice is rolled.

Solution: Let X_1 and X_2 be the random variables with $X_1((i, j)) = i$ and $X_2((i, j)) = j$, so that X_1 is the number appearing on the first die and X_2 is the number appearing on the second die.

It is easy to see that $E(X_1) = E(X_2) = 7/2$ because both equal $(1 + 2 + 3 + 4 + 5 + 6)/6 = 21/6 = 7/2$.

The sum of the two numbers that appear when the two dice are rolled is the sum $X_1 + X_2$. By Theorem 3, the expected value of the sum is $E(X_1 + X_2) = E(X_1) + E(X_2) = 7/2 + 7/2 = 7$.

7.4.6 Independent Random Variables

Definition 3: The random variables X and Y on a sample space S are **independent** if

$$p(X = r_1 \text{ and } Y = r_2) = p(X = r_1) \cdot p(Y = r_2),$$

or in words, if the probability that $X = r_1$ and $Y = r_2$ equals the product of the probabilities that $X = r_1$ and $Y = r_2$, for all real numbers r_1 and r_2 .

Theorem 5: If X and Y are independent variables on a sample space S , then $E(XY) = E(X)E(Y)$.

7.4.7 Variance₁

Deviation: The *deviation* (偏差, 离差) of X at $s \in S$ is $X(s) - E(X)$, the difference between the value of X and the mean of X .

Definition 4: Let X be a random variable on the sample space S . The *variance* of X , denoted by $V(X)$ is

$$V(X) = \sum_{s \in S} (X(s) - E(X))^2 p(s).$$

That is $V(X)$ is the weighted average of the square of the deviation of X . The *standard deviation* of X , denoted by $\sigma(X)$ is defined to be $\sqrt{V(X)}$.

Theorem 6: If X is a random variable on a sample space S , then
 $V(X) = E(X^2) - E(X)^2$.

Corollary 1: If X is a random variable on a sample space S and $E(X) = \mu$, then $V(X) = E((X - \mu)^2)$.

Variance₂

Example: What is the variance of the random variable X , where $X(t) = 1$ if a Bernoulli trial is a success and $X(t) = 0$ if it is a failure, where p is the probability of success and q is the probability of failure?

Solution: Because X takes only the values 0 and 1, it follows that $X^2(t) = X(t)$. Hence,

$$V(X) = E(X^2) - E(X)^2 = p - p^2 = p(1 - p) = pq.$$

Variance of the Value of a Die: What is the variance of a random variable X , where X is the number that comes up when a fair die is rolled?

Solution: We have $V(X) = E(X^2) - E(X)^2$. In an earlier example, we saw That $E(X) = 7/2$. Note that

$$E(X^2) = 1/6(1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2) = 91/6.$$

We conclude that $V(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{35}{12}$.

Variance₃

Irenée-Jules Bienaymé
(1796-1878)



Bienaymé's Formula: If X and Y are two independent random variables on a sample space S , then $V(X + Y) = V(X) + V(Y)$. Furthermore, if X_i , $i = 1, 2, \dots, n$, with n a positive integer, are pairwise independent random variables on S , then

$$V(X_1 + X_2 + \dots + X_n) = V(X_1) + V(X_2) + \dots + V(X_n).$$

Example: Find the **variance of the number of successes** when n independent Bernoulli trials are performed, where on each trial, p is the probability of success and q is the probability of failure.

Solution: Let X_i be the random variable with $X_i((t_1, t_2, \dots, t_n)) = 1$ if trial t_i is a success and $X_i((t_1, t_2, \dots, t_n)) = 0$ if trial t_i is a failure. Let $X = X_1 + X_2 + X_3 + \dots + X_n$. Then X counts the number of successes in the n trials.

- By Bienaymé's Formula, it follows that $V(X) = V(X_1) + V(X_2) + \dots + V(X_n)$.
- By the previous example, $V(X_i) = pq$ for $i = 1, 2, \dots, n$.

Hence, $V(X) = npq$.

Section Summary⁴

1, Expected Value $E(X) = \sum_{s \in S} p(s)X(s)$ $p(X = r) = \sum_{s \in S, X(s)=r} p(s)$, then

$$E(X) = \sum_{r \in X(S)} p(X = r)r.$$

2, Linearity of Expectations

(i) $E(X_1 + X_2 + \dots + X_n) = E(X_1) + E(X_2) + \dots + E(X_n)$

(ii) $E(aX + b) = aE(X) + b.$

3, Independent Random Variables

$$p(X = r_1 \text{ and } Y = r_2) = p(X = r_1) \cdot p(Y = r_2),$$

$$E(XY) = E(X)E(Y).$$

4, Variance $V(X) = \sum_{s \in S} (X(s) - E(X))^2 p(s).$ $V(X) = E((X - \mu)^2).$

$$V(X) = E(X^2) - E(X)^2.$$

$$V(X + Y) = V(X) + V(Y) \quad (X \text{ and } Y \text{ are two independent random variables})$$

Exercise

2. What is the expected number of heads that come up when a fair coin is flipped 10 times?
3. What is the expected number of times a 6 appears when a fair die is rolled 10 times?
27. What is the variance of the number of heads that come up when a fair coin is flipped 10 times?

Homework

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